GAP-10D: Conditional Low-Energy Uniqueness of Einstein Dynamics

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Abstract

The covariant-tetrad architecture does not by itself derive the Einstein equations. This note closes two conditional subproblems. First, variation of the minimal Hilbert–Palatini action with respect to the tetrad and independent connection gives the Einstein equation and the Cartan torsion equation. Second, Lovelock uniqueness implies that in four dimensions any local natural symmetric divergence-free metric equation of second differential order has the form $aG_{\mu\nu} + bg_{\mu\nu}$. Thus, once UBT derives the standard low-energy assumptions, the classical metric equation is fixed to Einstein with a cosmological constant. The remaining fundamental gap is the derivation of those assumptions, coefficients and matter coupling from the canonical single-Θ action, not ambiguity among many second-order metric equations.

1 Minimal first-order action

Use the action

$$S[e,\omega,\Theta] = \frac{1}{4\kappa}\int \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}(\omega) - \frac{\Lambda}{24\kappa}\int \epsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d + S_{\mathrm{m}}[e,\omega,\Theta]. \quad (1)$$

Define the energy-momentum three-form by $\delta_e S_{\mathrm{m}} = \int \Sigma_a \wedge \delta e^a$ and spin current as in the companion torsion note.

Theorem 1.1 (First-order field equations)**.** *Independent variation of Eq.* (1) *gives*

$$\epsilon_{abcd}\, e^b \wedge R^{cd} - \frac{\Lambda}{3}\epsilon_{abcd}\, e^b \wedge e^c \wedge e^d = -2\kappa\, \Sigma_a, \quad (2)$$

$$\epsilon_{abcd}\, T^a \wedge e^b = \kappa \tau_{cd}. \quad (3)$$

When $\tau_{ab} = 0$, the second equation gives the Levi–Civita connection and the first is equivalent to

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.} \quad (4)$$

Remark 1.2. This is standard tetrad/Palatini gravity. It proves what follows from the minimal action; it does not prove that Eq. (1) has already been derived from the deeper UBT invariant.

2 Four-dimensional uniqueness

Theorem 2.1 (Lovelock consequence in four dimensions). *Assume the classical metric equation is a local natural symmetric rank-two tensor* $\mathcal{E}_{\mu\nu}[g]$ *which*

1. *depends on the metric and at most its first two derivatives;*
2. *is divergence free identically;*
3. *contains no additional propagating geometric fields in the branch;*
4. *is of the standard second-order/Lovelock class.*

Then in four dimensions

$$\boxed{\mathcal{E}_{\mu\nu} = aG_{\mu\nu} + bg_{\mu\nu}.} \tag{5}$$

After fixing normalization and writing $\Lambda = b/a$, *the sourced equation is Einstein's equation with cosmological constant.*

Corollary 2.2 (UBT low-energy selection). *If the on-shell UBT tetrad branch is local, diffeomorphism and local-Lorentz invariant, torsion free at macroscopic scales, second order, and contains no additional light geometric fields, then its metric dynamics cannot be an arbitrary alternative tensor equation: it is Einstein–*Λ *up to the overall coupling.*

3 What this does not derive

The theorem does not determine from UBT:

- why the effective action is local and in the Lovelock class;
- why higher-curvature or nonlocal terms are absent or suppressed;
- the values and signs of κ and Λ;
- the exact matter stress tensor and paired spin current;
- the on-shell solution of $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$.

4 Status

Defensible status

GAP-10D-PALATINI: CLOSED CONDITIONALLY [L1]. The minimal first-order tetrad action yields the Cartan equation and Einstein–Λ dynamics.

GAP-10D-UNIQUENESS: CLOSED CONDITIONALLY [L1]. Under the four-dimensional Lovelock/low-energy assumptions, the only metric equation is $aG_{\mu\nu} + bg_{\mu\nu} = \kappa T_{\mu\nu}$.

GAP-10D: NARROWED, NOT FULLY CLOSED. The remaining UBT task is to derive the minimal low-energy action or the Lovelock hypotheses, its coefficients and matter terms from the canonical single-field theory. The conditional infrared endpoint is no longer ambiguous.

References

[1] D. Lovelock, “The Einstein Tensor and Its Generalizations,” *J. Math. Phys.* **12**, 498–501 (1971).

[2] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, “General Relativity with Spin and Torsion: Foundations and Prospects,” *Rev. Mod. Phys.* **48**, 393–416 (1976).